DANIEL MORA

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EDUCATION

IE University **Madrid, Spain**
Bachelor in Computer Science and Artificial Intelligence 2026

EXPERIENCE

WhiteHole (hedge fund) **San Sebastián, Spain**
Quantitative Research Intern Jun 2025 – Nov 2025

- Forecast CPI and GDP from housing-market indicators, from feature engineering to out-of-sample validation.
- Engineered NLP features from public political communications and tested their power to predict market volatility in tree-based, linear and ridge regression models, including regime shifts.
- Built a knowledge graph from text using NLP extraction and locally hosted language models.

MAI-Book **Remote**
Front-end Developer Intern Jun 2023 – Oct 2023

- Developed user-facing web interfaces for an online library, optimized for engagement and conversion.
- Ran A/B tests and UX research to improve usability and retention across the sales funnel.

RESEARCH AND PROJECTS

JEPA-Embedding Transformer (ArgusJEPA + CassaFormer) PyTorch, Triton 2026

- Causal Transformer forecasting next-day US equity targets from JEPA embedding sequences; code and weights public.
- ArgusJEPA, a self-supervised encoder (EMA teacher, 50% masked views), maps each stock-day to a 256-d embedding; frozen, these feed CassaFormer, a 2.05M-parameter decoder with RoPE, QK-RMSNorm, top-1 MoE and a Student-t head.
- Built the point-in-time data platform behind it (IBKR, SEC filings by acceptance time, FRED/ALFRED vintages, scored news) on PostgreSQL and DuckDB over a survivorship-free top-500 universe.
- Fused Triton kernels kept only where they beat `torch.compile` (L4): MoE 1.2–4.1$\times$ per layer, 1.1–1.5$\times$ end to end.

TC_ACC: Autonomous Documentary Studio Python, LangGraph, Remotion 2026

- Agentic system turning a case reference into a narrated 10-minute documentary in about five hours, unattended.
- 29 stages, 6 departments, 2 review gates; typed Pydantic artifacts, resumable stages, 46 CI tests run keyless.

Bachelor Thesis (draft paper, ICLR format) Supervisor: Dr. Luciano Dyballa

- “A Cross-Description Test of Sparse Feature Correspondence Between Language Models and fMRI Responses.”
- Tested whether brain and LM mappings share sparse features: 59 participants, six SAE systems (1B–12B).

Sparse Autoencoder Releases PyTorch, Hugging Face, GCP

- OLMo SAE Builder: released seven SAELens-compatible TopK SAEs for OLMo-3-7B-Think.
- Triplet SAE Project: SAE steering vs. LoRA for triplet extraction; shipped a fine-tuned Gemma 3 4B (GPTQ).

LEADERSHIP

- **Founder**, IE Quant Finance Club.
- **Founder and President**, IE Dominican Republic Club: IE’s largest student-led diversity event (500+).
- **Business Development Officer**, IE Investment Club: secured sponsorships with financial-sector firms.

SKILLS

ML and research: PyTorch, Triton, Hugging Face (Transformers, PEFT/LoRA, SAELens), JEPA, MoE, NLP
Engineering: Python, C++, SQL; LangGraph, Pydantic, FastAPI; PostgreSQL, DuckDB; Docker, GCP, Azure
Languages: Spanish (native), English (fluent)